```
{
    stream << ob.feet << " feet is " << ob.inches;
    stream << " inches\n";

    return stream;
}

int main()
{
    ft_to_inches x;

    cin >> x;
    cout << x;

    return 0;
}
```

REVIEW SKILLS CHECK: Chapter 9

1.
```
#include <iostream>
using namespace std;

int main()
{
    cout.width(40);
    cout.fill(':');

    cout << "C++ is fun" << '\n';

    return 0;
}
```

2.
```
#include <iostream>
using namespace std;

int main()
{
    cout.precision(4);
    cout << 10.0/3.0 << '\n';

    return 0;
}
```

3.
```
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    cout << setprecision(4) << 10.0/3.0 << '\n';
```

```
    return 0;
}
```

4. An inserter is an overloaded **operator<<()** function that outputs a class's data to an output stream. An extractor is an overloaded **operator>>()** function that inputs a class's data from an input stream.

5. ```
#include <iostream>
using namespace std;

class date
{
    char d[9]; // store date as string: mm/dd/yy
public:
    friend ostream &operator<<(ostream &stream, date ob);
    friend istream &operator>>(istream &stream, date &ob);
};

ostream &operator<<(ostream &stream, date ob)
{
    stream << ob.d << '\n';

    return stream;
}

istream &operator>>(istream &stream, date &ob)
{
    cout << "Enter date (mm/dd/yy): ";
    stream >> ob.d;

    return stream;
}

int main()
{
    date ob;

    cin >> ob;
    cout << ob;

    return 0;
}
```

6. To use a parameterized manipulator, you must include **<iomanip>** in your program.

7. The predefined streams are **cin**, **cout**, **cerr**, and **clog**.

## 9.1 EXERCISES

1. ```
// Show time and date.
```

```
#include <iostream>
#include <ctime>
using namespace std;

// A time and date output manipulator.
ostream &td(ostream &stream)
{
    struct tm *localt;
    time_t t;

    t = time(NULL);
    localt = localtime(&t);
    stream << asctime(localt) << endl;

    return stream;
}

int main()
{
    cout << td << '\n';

    return 0;
}
```

2.
```
#include <iostream>
using namespace std;

// Turn on hex output with uppercase X.
ostream &sethex(ostream &stream)
{
    stream.unsetf(ios::dec | ios::oct);
    stream.setf(ios::hex | ios::uppercase |
                ios::showbase);

    return stream;
}

// Reset flags.
ostream &reset(ostream &stream)
{
    stream.unsetf(ios::hex | ios::uppercase |
                ios::showbase);
    stream.setf(ios::dec);

    return stream;
}

int main()
{
    cout << sethex << 100 << '\n';
    cout << reset << 100 << '\n';
```

```
    return 0;
}
```

3.
```
#include <iostream>
using namespace std;

// Skip 10 characters.
istream &skipchar(istream &stream)
{
    int i;
    char c;

    for(i=0; i<10; i++)
        stream >> c;

    return stream;
}

int main()
{
    char str[80];

    cout << "Enter some characters: ";
    cin >> skipchar >> str;

    cout << str << '\n';

    return 0;
}
```

9.2 EXERCISES

1.
```
// Copy a text file and display number of chars copied.
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=3)
    {
        cout << "Usage: CPY <input> <output>\n";
        return 1;
    }

    ifstream fin(argv[1]); // open input file.
    ofstream fout(argv[2]); // create output file

    if(!fin)
```

```
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    if(!fout)
    {
        cout << "Cannot open output file.\n";
        return 1;
    }

    char ch;
    unsigned count = 0;

    fin.unsetf(ios::skipws); // do not skip spaces
    while(!fin.eof())
    {
        fin >> ch;
        if(!fin.eof())
        {
            fout << ch;
            count++;
        }
    }

    cout << "Number of bytes copied: " << count << '\n';

    fin.close();
    fout.close();

    return 0;
}
```

The reason this program might display a result different from that shown when you list the directory is that some character translations might be taking place. Specifically, when a carriage-return/linefeed sequence is read, it is converted into a newline. When output, newlines are counted as one character but converted back into a carriage-return/linefeed sequence again.

2.
```
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    ofstream pout("phone");

    if(!pout)
    {
        cout << "Cannot open PHONE file.\n";
        return 1;
    }
```

```
    pout << "Isaac Newton 415 555-3423\n";
    pout << "Robert Goddard 213 555-2312\n";
    pout << "Enrico Fermi 202 555-1111\n";

    pout.close();

    return 0;
}
```

3.
```
// Word count
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: COUNT <input>\n";
        return 1;
    }

    ifstream in(argv[1]);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    int count = 0;
    char ch;

    in >> ch; // find first non-space char

    // after first non-space found, do not skip spaces
    in.unsetf(ios::skipws); // do not skip spaces

    while(!in.eof())
    {
        in >> ch;
        if(isspace(ch))
        {
            count++;
            while(isspace(ch) && !in.eof())
                in >> ch;
        }
    }
```

```
    cout << "Word count: " << count << '\n';

    in.close();

    return 0;
}
```

4. The **is_open()** function returns true if the invoking stream is linked to an open file.

9.3 EXERCISES

1a.
```
// Copy a file and display number of chars copied.
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=3)
    {
        cout << "Usage: CPY <input> <output>\n";
        return 1;
    }

    ifstream fin(argv[1], ios::in | ios::binary); // open
       input file
    ofstream fout(argv[2], ios::out | ios::binary); // create
        output file

    if(!fin)
    {
        cout << "Cannot open input file\n";
        return 1;
    }

    if(!fout)
    {
        cout << "Cannot open output file\n";
        return 1;
    }

    char ch;
    unsigned count=0;

    while(!fin.eof())
    {
        fin.get(ch);
        if(!fin.eof())
        {
            fout.put(ch);
            count++;
```

```
        }
    }

    cout << "Number of bytes copied: " << count << '\n';

    fin.close();
    fout.close();

    return 0;
}
```

1b.
```
// Word count.
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: COUNT <input>\n";
        return 1;
    }

    ifstream in(argv[1], ios::in | ios::binary);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    int count = 0;
    char ch;

    // find first non-space char
    do
    {
        in.get(ch);
    }
    while(isspace(ch));

    while(!in.eof())
    {
        in.get(ch);
        if(isspace(ch))
        {
            count++;
            while(isspace(ch) && !in.eof())
                in.get(ch); // find next word
```

```
        }
    }

    cout << "Word count: " << count << '\n';

    in.close();

    return 0;
}
```

2.
```
// Output account info to a file using an inserter.
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

class account
{
    int custnum;
    char name[80];
    double balance;
public:
    account(int c, char *n, double b)
    {
        custnum = c;
        strcpy(name, n);
        balance = b;
    }
    friend ostream &operator<<(ostream &stream, account ob);
};

ostream &operator<<(ostream &stream, account ob)
{
    stream << ob.custnum << ' ';
    stream << ob.name << ' ' << ob.balance;
    stream << '\n';

    return stream;
}

int main()
{
    account Rex(1011, "Ralph Rex", 12323.34);
    ofstream out("accounts", ios::out | ios::binary);

    if(!out)
    {
        cout << "Cannot open output file.\n";
        return 1;
    }
```

```
    out << Rex;

    out.close();

    return 0;
}
```

9.4 EXERCISES

1.
```
// Use get() to read a string that contains spaces.
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    char str[80];

    cout << "Enter your name: ";
    cin.get(str, 79);

    cout << str << '\n';

    return 0;
}
```

   The program functions the same whether it uses **get()** or **getline()**.

2.
```
// Use getline() to display a text file.
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: PR <filename>\n";
        return 1;
    }

    ifstream in(argv[1]);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    char str[255];
```

```
    while(!in.eof())
    {
        in.getline(str, 254);
        cout << str << '\n';
    }

    in.close();

    return 0;
}
```

9.5 EXERCISES

1.
```
// Display a file backwards on the screen.
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: REVERSE <filename>\n";
        return 1;
    }

    ifstream in(argv[1], ios::in | ios::binary);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    char ch;
    long i;

    // go to end of file (less eof char)
    in.seekg(0, ios::end);
    i = (long) in.tellg(); // see how many bytes in file
    i -= 2; // backup before eof

    for( ; i>=0; i--)
    {
        in.seekg(i, ios::beg);
        in.get(ch);
        cout << ch;
    }

    in.close();
```

```
    return 0;
}
```

2.
```
// Swap characters in a file.
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: SWAP <filename>\n";
        return 1;
    }

    // opne file for input/output
    fstream io(argv[1], ios::in | ios::out | ios::binary);
    if(!io)
    {
        cout << "Cannot open file.\n";
        return 1;
    }

    char ch1, ch2;
    long i;

    for(i=0; !io.eof(); i+=2)
    {
        io.seekg(i, ios::beg);
        io.get(ch1);
        if(io.eof())
            continue;
        io.get(ch2);
        if(!io.eof())
            continue;
        io.seekg(i, ios::beg);
        io.put(ch2);
        io.put(ch1);
    }

    io.close();

    return 0;
}
```

9.6 EXERCISES

1a.
```
/*
   Display a file backwards on the screen,
```

```
   plus error checking.
*/
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: REVERSE <filename>\n";
        return 1;
    }

    ifstream in(argv[1], ios::in | ios::binary);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    char ch;
    long i;

    // go to end of file (less eof char)
    in.seekg(0, ios::end);
    if(!in.good())
        return 1;
    i = (long) in.tellg(); // see how many bytes in file
    if(!in.good())
        return 1;
    i -= 2; // backup before eof

    for( ; i>=0; i--)
    {
        in.seekg(i, ios::beg);
        if(!in.good())
            return 1;
        in.get(ch);
        if(!in.good())
            return 1;
        cout << ch;
    }

    in.close();
    if(!io.good())
        return 1;

    return 0;
}
```

1b.
```
// Swap characters in a file with error checking.
#include <iostream>
#include <fstream>
using namespace std;

int main(int argc, char *argv[])
{
    if(argc!=2)
    {
        cout << "Usage: SWAP <filename>\n";
        return 1;
    }

    // opne file for input/output
    fstream io(argv[1], ios::in | ios::out | ios::binary);
    if(!io)
    {
        cout << "Cannot open file.\n";
        return 1;
    }

    char ch1, ch2;
    long i;

    for(i=0; !io.eof(); i+=2)
    {
        io.seekg(i, ios::beg);
        if(!io.good())
            return 1;
        io.get(ch1);
        if(io.eof())
            continue;
        io.get(ch2);
        if(!io.good())
            return 1;
        if(!io.eof())
            continue;
        io.seekg(i, ios::beg);
        if(!io.good())
            return 1;
        io.put(ch2);
        if(!io.good())
            return 1;
        io.put(ch1);
        if(!io.good())
            return 1;
    }

    io.close();
    if(!io.good())
        return 1;
```

```
    return 0;
}
```

MASTERY SKILLS CHECK: Chapter 9

1.
```
#include <iostream>
using namespace std;

ostream &tabs(ostream &stream)
{
    stream << '\t' << '\t' << '\t';
    stream.width(20);

    return stream;
}

int main()
{
    cout << tabs << "Testing\n";

    return 0;
}
```

2.
```
#include <iostream>
#include <cctype>
using namespace std;

istream &findalpha(istream &stream)
{
    char ch;

    do
    {
        stream.get(ch);
    }
    while(!isalpha(ch));

    return stream;
}

int main()
{
    char str[80];

    cin >> findalpha >> str;
    cout << str << '\n';

    return 0;
}
```

3.
```
// Copy a file and reverse case of letters.
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

int main(int argc, char *argv[])
{
    char ch;

    if(argc!=3)
    {
        cout << "Usage: COPYREV <source> <target>\n";
        return 1;
    }

    ifstream in(argv[1]);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    ofstream out(argv[2]);

    if(!out)
    {
        cout << "Cannot open output file.\n";
        return 1;
    }

    while(!in.eof())
    {
        ch = in.get();
        if(!in.eof())
        {
            if(islower(ch))
                ch = toupper(ch);
            else
                ch = tolower(ch);
            out.put(ch);
        }
    }

    in.close();
    out.close();

    return 0;
}
```

4.
```
// Count letters.
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

int alpha[26];

int main(int argc, char *argv[])
{
    char ch;

    if(argc!=2)
    {
        cout << "Usage: COUNT <source>\n";
        return 1;
    }

    ifstream in(argv[1]);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    // init alpha[]
    int i;
    for(i=0; i<26; i++)
        alpha[i] = 0;

    while(!in.eof())
    {
        ch = in.get();
        if(isalpha(ch)) // if letter found, count it
        {
            ch = toupper(ch); // normalize
            alpha[ch-'A']++; // 'A'-'A' == 0, 'B'-'A' == 1,
               etc.
        }
    }

    // display count
    for(i=0; i<26; i++)
        cout << (char) ('A'+i) << ": " << alpha[i] << '\n';

    in.close();

    return 0;
}
```

5a.
```
/*
   Copy a file and reverse case of letters
   with error checking.
*/
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

int main(int argc, char *argv[])
{
    char ch;

    if(argc!=3)
    {
        cout << "Usage: COPYREV <source> <target>\n";
        return 1;
    }

    ifstream in(argv[1]);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    ofstream out(argv[2]);

    if(!out)
    {
        cout << "Cannot open output file.\n";
        return 1;
    }

    while(!in.eof())
    {
        ch = in.get();
        if(!in.good() && !in.eof())
            return 1;
        if(!in.eof())
        {
            if(islower(ch))
                ch = toupper(ch);
            else
                ch = tolower(ch);
            out.put(ch);
        if(!out.good())
            return 1;
        }
    }
```

```
    in.close();
    out.close();
    if(!in.good() && !out.good())
        return 1;

    return 0;
}
```

5b.
```
// Count letters with error checking.
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

int alpha[26];

int main(int argc, char *argv[])
{
    char ch;

    if(argc!=2)
    {
        cout << "Usage: COUNT <source>\n";
        return 1;
    }

    ifstream in(argv[1]);

    if(!in)
    {
        cout << "Cannot open input file.\n";
        return 1;
    }

    // init alpha[]
    int i;
    for(i=0; i<26; i++)
        alpha[i] = 0;

    while(!in.eof())
    {
        ch = in.get();
        if(!in.good() && !in.eof())
            return 1;
        if(isalpha(ch)) // if letter found, count it
        {
            ch = toupper(ch); // normalize
            alpha[ch-'A']++; // 'A'-'A' == 0, 'B'-'A' == 1,
               etc.
        }
```

```
    }

    // display count
    for(i=0; i<26; i++)
        cout << (char) ('A'+i) << ": " << alpha[i] << '\n';

    in.close();
    if(!in.good())
        return 1;

    return 0;
}
```

6. To set the **get** pointer, use **seekg()**. To set the **put** pointer, use **seekp()**.

CUMULATIVE SKILLS CHECK: Chapter 9

1.
```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

#define SIZE 40

class inventory
{
    char item[SIZE]; // name of item
    int onhand; // number on hand
    double cost; // cost of item
public:
    inventory(char *i, int o, double c)
    {
        strcpy(item, i);
        onhand = o;
        cost = c;
    }
    void store(fstream &stream);
    void retrieve(fstream &stream);
    friend ostream &operator<<(ostream &stream, inventory ob)
       ;
    friend istream &operator>>(istream &stream, inventory &ob
       );
};

ostream &operator<<(ostream &stream, inventory ob)
{
    stream << ob.item << ": " << ob.onhand;
    stream << " on hand at $" << ob.cost << '\n';

    return stream;
}
```

```
istream &operator>>(istream &stream, inventory &ob)
{
    cout << "Enter item name: ";
    stream >> ob.item;
    cout << "Enter number on hand: ";
    stream >> ob.onhand;
    cout << "Enter cost: ";
    stream >> ob.cost;

    return stream;
}

void inventory::store(fstream &stream)
{
    stream.write(item, SIZE);
    stream.write((char *) &onhand, sizeof(int));
    stream.write((char *) &cost, sizeof(double));
}

void inventory::retrieve(fstream &stream)
{
    stream.read(item, SIZE);
    stream.read((char *) &onhand, sizeof(int));
    stream.read((char *) &cost, sizeof(double));
}

int main()
{
    fstream inv("inv", ios::out | ios::binary);
    int i;

    inventory pliers("pliers", 12, 4.95);
    inventory hammers("hammers", 5, 9.45);
    inventory wrenches("wrenches", 22, 13.90);
    inventory temp("", 0, 0.0);

    if(!inv)
    {
        cout << "Cannot open file for output.\n";
        return 1;
    }

    // write to file
    pliers.store(inv);
    hammers.store(inv);
    wrenches.store(inv);

    inv.close();

    // open for input
```

```
    inv.open("inv", ios::in | ios::binary);

    if(!inv)
    {
        cout << "Cannot open file for input.\n";
        return 1;
    }

    do
    {
        cout << "Record # (-1 to quit): ";
        cin >> i;
        if(i == -1)
            break;
        inv.seekg(i*(SIZE+sizeof(int)+sizeof(double)), ios::
           beg);
        temp.retrieve(inv);
        cout << temp;
    }
    while(inv.good());

    inv.close();

    return 0;
}
```

REVIEW SKILLS CHECK: Chapter 10

1. ```
   #include <iostream>
   using namespace std;

   ostream &setsci(ostream &stream)
   {
       stream.setf(ios::scientific | ios::uppercase);

       return stream;
   }

   int main()
   {
       double f = 123.23;

       cout << setsci << f;
       cout << '\n';

       return 0;
   }
   ```

2. ```
   // Copy and convert tabs to spaces.
   #include <iostream>
   ```